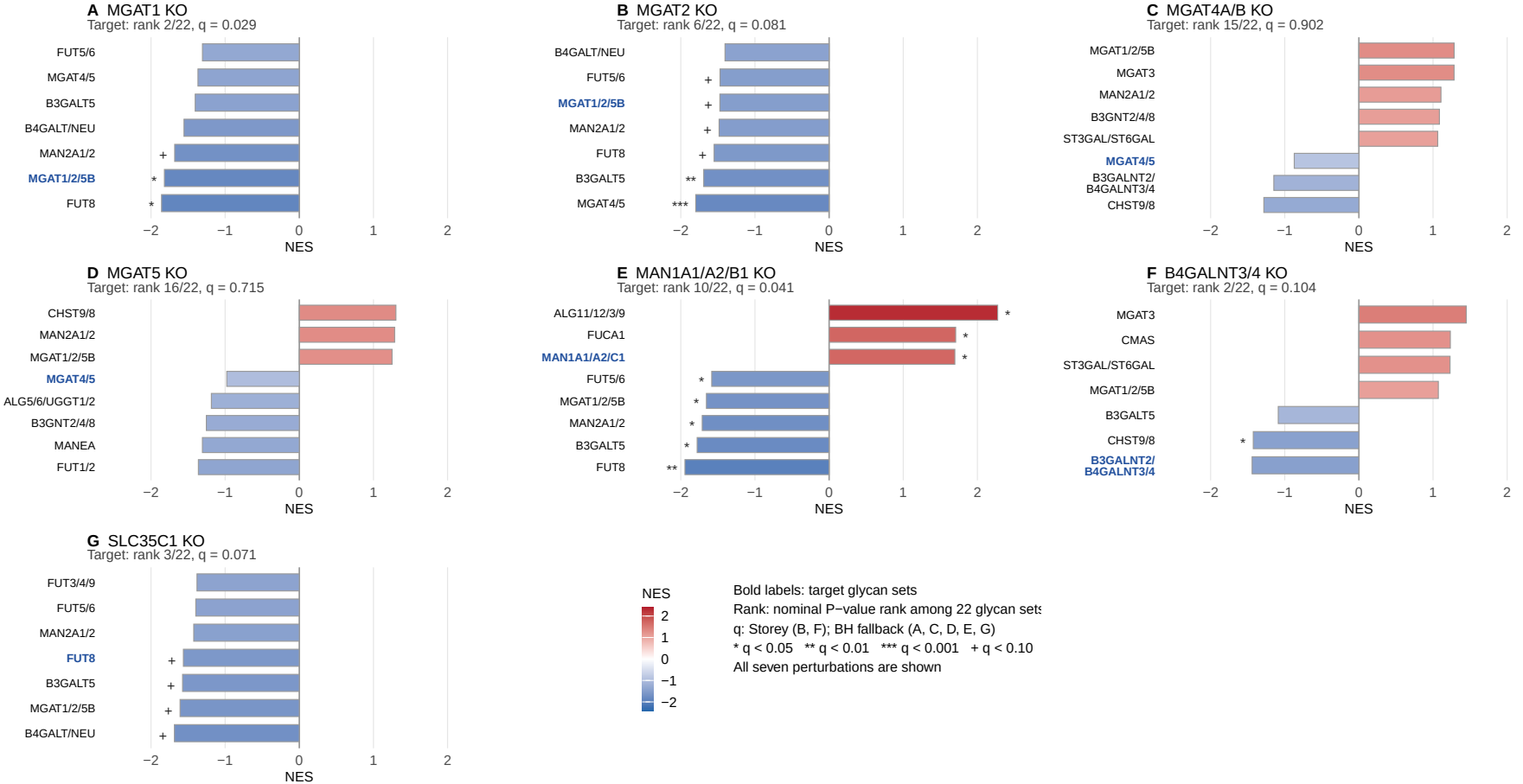
A MGAT1 KO
Target: rank 2/22, q = 0.029
FUT5/6
MGAT4/5
B3GALT5
B4GALT/NEU
MAN2A1/2
MGAT1/2/5B
FUT8
+
*
*
−2
−1
0
1
2
NES
B MGAT2 KO
Target: rank 6/22, q = 0.081
B4GALT/NEU
FUT5/6
MGAT1/2/5B
MAN2A1/2
FUT8
B3GALT5
MGAT4/5
+
+
+
+
**
***
−2
−1
0
1
2
NES
C MGAT4A/B KO
Target: rank 15/22, q = 0.902
MGAT1/2/5B
MGAT3
MAN2A1/2
B3GNT2/4/8
ST3GAL/ST6GAL
MGAT4/5
B3GALNT2/
B4GALNT3/4
CHST9/8
−2
−1
0
1
2
NES
D MGAT5 KO
Target: rank 16/22, q = 0.715
CHST9/8
MAN2A1/2
MGAT1/2/5B
MGAT4/5
ALG5/6/UGGT1/2
B3GNT2/4/8
MANEA
FUT1/2
−2
−1
0
1
2
NES
E MAN1A1/A2/B1 KO
Target: rank 10/22, q = 0.041
ALG11/12/3/9
FUCA1
MAN1A1/A2/C1
FUT5/6
MGAT1/2/5B
MAN2A1/2
B3GALT5
FUT8
*
*
*
*
*
*
**
−2
−1
0
1
2
NES
F B4GALNT3/4 KO
Target: rank 2/22, q = 0.104
MGAT3
CMAS
ST3GAL/ST6GAL
MGAT1/2/5B
B3GALT5
CHST9/8
B3GALNT2/
B4GALNT3/4
*
−2
−1
0
1
2
NES
G SLC35C1 KO
Target: rank 3/22, q = 0.071
FUT3/4/9
FUT5/6
MAN2A1/2
FUT8
B3GALT5
MGAT1/2/5B
B4GALT/NEU
+
+
+
+
−2
−1
0
1
2
NES
NES
2
1
0
−1
−2
Bold labels: target glycan sets
Rank: nominal P−value rank among 22 glycan sets
q: Storey (B, F); BH fallback (A, C, D, E, G)
* q < 0.05 ** q < 0.01 *** q < 0.001 + q < 0.10
All seven perturbations are shown